SAFARI AUTO SPARES

Microsoft Access Practical Test — Complete Answer & Step-by-Step Guide

Based strictly on the uploaded Form Four ICS Database Management System practical paper.

DATABASE STRUCTURE

Create a blank Microsoft Access database and save it as **SAFARI AUTO SPARES.accdb**. The paper uses SUPPLIERS, PARTS and ORDERS.

Field	Data Type	Key / Property
SupplierID	Short Text	Primary Key
SupplierName	Short Text	
Region	Short Text	
PhoneNumber	Short Text	
ContractStartDate	Date/Time	

Field	Data Type	Key / Property
PartCode	Short Text	Primary Key
PartName	Short Text	
Category	Short Text	Validation later
UnitPrice	Currency	
QtyInStock	Number (Long Integer)	Validation later
SupplierID	Short Text	Foreign Key
DateReceived	Date/Time	

Field	Data Type	Key / Property
OrderID	Short Text	Primary Key
PartCode	Short Text	Foreign Key
GarageName	Short Text	
OrderDate	Date/Time	
QtyOrdered	Number (Long Integer)	
DeliveryStatus	Short Text	Lookup later

PART (a) — Create database, tables and records

• Open Microsoft Access → Blank Database → enter SAFARI AUTO SPARES.accdb → Create.

• Create each table using Create → Table Design. Use the field names and data types in the tables above.

• Set SupplierID, PartCode and OrderID as the primary keys.

• Save the tables exactly as SUPPLIERS, PARTS and ORDERS.

• Switch to Datasheet View and enter every record from the question.

• Use Short Text for PhoneNumber so the leading zero is preserved; use Date/Time for dates.

Source data: 7 suppliers, 12 parts and 10 orders.

PART (b) — Primary and foreign keys

• SUPPLIERS: SupplierID = Primary Key.

• PARTS: PartCode = Primary Key; SupplierID = Foreign Key.

• ORDERS: OrderID = Primary Key; PartCode = Foreign Key.

• Foreign-key fields must use compatible data types with their parent primary keys.

PART (c) — Relationships

• Go to Database Tools → Relationships → Add the three tables.

• Drag SUPPLIERS.SupplierID to PARTS.SupplierID. Tick Enforce Referential Integrity and click Create.

• This is a one-to-many relationship: SUPPLIERS (1) → (∞) PARTS.

• Drag PARTS.PartCode to ORDERS.PartCode. Tick Enforce Referential Integrity and click Create.

• This is a one-to-many relationship: PARTS (1) → (∞) ORDERS.

• Save the Relationships layout.

PART (d) — Field controls

• **DeliveryStatus:** Open ORDERS in Design View → select DeliveryStatus → Lookup Wizard → “I will type in the values that I want” → enter Pending, Delivered and Cancelled on separate rows → Finish → Save.

• **QtyInStock:** Open PARTS in Design View → select QtyInStock → Validation Rule: >=0 → Validation Text: Quantity cannot be negative → Save.

• **Category:** select Category → Validation Rule: In ("Engine","Body","Electrical","Suspension") → Validation Text: Invalid category → Save.

PART (e) — QUERIES

For each query: Create → Query Design → add the required tables → add fields/criteria, or switch to SQL View → paste the SQL → Run (!) → Save with the exact query name.

Critical Stock

```
SELECT * FROM PARTS
WHERE UnitPrice > 50000
AND QtyInStock < 10;
```

PartCode	PartName	UnitPrice	QtyInStock
P102	Alternator	120,000	5
P104	Radiator	150,000	6
P107	Bonnet Cover	175,000	3
P108	Fuel Injector	210,000	4
P111	Engine Mounting	54,000	7

BE Parts

```
SELECT * FROM PARTS
WHERE (PartName Like "B*" OR PartName Like "*r")
AND Category In ("Engine","Electrical");
```

PartCode	PartName	Category
P102	Alternator	Electrical
P104	Radiator	Engine
P105	Battery Terminal	Electrical
P108	Fuel Injector	Engine
P110	Headlamp Cover	Electrical

Stock Valuation

```
SELECT PARTS.*, [UnitPrice]*[QtyInStock] AS TotalValue,
([UnitPrice]*[QtyInStock])*0.18 AS VAT
FROM PARTS;
```

PartCode	PartName	TotalValue	VAT
P101	Brake Pad	360,000	64,800
P102	Alternator	600,000	108,000
P103	Bumper Set	1,330,000	239,400
P104	Radiator	900,000	162,000
P105	Battery Terminal	320,000	57,600
P106	Ball Joint	342,000	61,560
P107	Bonnet Cover	525,000	94,500
P108	Fuel Injector	840,000	151,200
P109	Shock Absorber	737,000	132,660
P110	Headlamp Cover	442,000	79,560
P111	Engine Mounting	378,000	68,040
P112	Bonnet Lock	330,000	59,400

Supply Records

```
SELECT SUPPLIERS.SupplierName, SUPPLIERS.Region,
PARTS.PartName, PARTS.UnitPrice
```

```
FROM SUPPLIERS INNER JOIN PARTS
ON SUPPLIERS.SupplierID = PARTS.SupplierID
WHERE PARTS.DateReceived Between #03/01/2023# AND #04/30/2023#;
```

SupplierName	Region	PartName	UnitPrice
Ruaha Engineering	Iringa	Radiator	150,000
Twiga Auto Supply	Arusha	Battery Terminal	8,000
Serengeti Motors	Mwanza	Ball Joint	38,000
Coastal Spares Ltd	Tanga	Bonnet Cover	175,000
Kilimanjaro Motors	Arusha	Fuel Injector	210,000
Bahari Auto Parts	Dar es Salaam	Shock Absorber	67,000

Mid Range Parts

```
SELECT * FROM PARTS
WHERE UnitPrice BETWEEN 20000 AND 100000
AND Category <> "Body";
```

PartCode	PartName	Category	UnitPrice
P101	Brake Pad	Suspension	45,000
P106	Ball Joint	Suspension	38,000
P109	Shock Absorber	Suspension	67,000
P110	Headlamp Cover	Electrical	26,000
P111	Engine Mounting	Engine	54,000

Pending Arusha Orders

```
SELECT ORDERS.GarageName, PARTS.PartName,
SUPPLIERS.Region, ORDERS.OrderDate
FROM (SUPPLIERS INNER JOIN PARTS
ON SUPPLIERS.SupplierID = PARTS.SupplierID)
INNER JOIN ORDERS ON PARTS.PartCode = ORDERS.PartCode
WHERE ORDERS.DeliveryStatus = "Pending"
AND SUPPLIERS.Region = "Arusha";
```

GarageName	PartName	Region	OrderDate
Furaha Motors	Fuel Injector	Arusha	15/04/2023
Neema Garage	Fuel Injector	Arusha	18/04/2023

Popular Parts

```
SELECT PartCode, Sum(QtyOrdered) AS TotalQtyOrdered
FROM ORDERS
GROUP BY PartCode
HAVING Count(*) > 1;
```

PartCode	TotalQtyOrdered
P101	21
P104	7
P108	3

PART (f) — FORMS

• Go to Create → Form Wizard.

• For SUPPLIERS, select all fields → choose a suitable layout such as Columnar → Finish → save as SupplierForm.

• Repeat for PARTS and save as PartForm.

• Repeat for ORDERS and save as OrderForm.

• Open each form and verify that all records are present and correctly entered.

PART (g) — CATEGORY REPORT

• Go to Create → Report Wizard.

• Use PARTS (or a suitable query containing TotalValue) as the report source.

• Select PartName, UnitPrice and QtyInStock, and make TotalValue available as UnitPrice × QtyInStock.

• Group the report by Category.

• Set the group summary to Sum of TotalValue.

• Finish and save the report as Category Report.

• Open Design/Layout View, add a Page Header date control and set its Control Source to =Date().

• Preview the report and check that every category shows its parts and the category TotalValue sum.

Expected category totals:

Category	Sum of TotalValue (Tsh)
Body	2,185,000
Electrical	1,362,000
Engine	2,118,000
Suspension	1,439,000

FINAL PRINTING CHECKLIST

• Database name: SAFARI AUTO SPARES.

• Tables: SUPPLIERS, PARTS, ORDERS.

• Primary keys and both foreign keys are correct.

• Both relationships have Enforce Referential Integrity enabled.

• DeliveryStatus has the three-value drop-down.

• QtyInStock rejects negative numbers with “Quantity cannot be negative”.

• Category rejects invalid entries with “Invalid category”.

• All seven queries are saved with the exact required names.

• Forms are saved as SupplierForm, PartForm and OrderForm.

• Report is saved as Category Report, grouped by Category, with current date in the header.

• Print all required work: tables, relationships, queries, forms and report.